

Visitor Management Backend - Technical Documentation

Spring Boot 4.1.0 · Java 21 · PostgreSQL 17 · Hibernate 7.4

1. Architecture Overview

+-----+ REST Controllers +-----+	
TimeSlotController · VisitorController	
RegistrationController · BriefingController	
EventController	
+-----+ Services +-----+	
SchedulingServiceImpl · RegistrationServiceImpl	
EventServiceImpl	
SchedulingRules (constants)	
+-----+ Repositories (JPA) +-----+	
TimeSlotRepository · VisitorRepository	
RegistrationRepository · BriefingRepository	
EventRepository	
+-----+ JPA Entities +-----+	
TimeSlot · Visitor · Registration · Briefing · Event	
(extend BaseEntity: id + createdAt + updatedAt)	
+-----+ core/ - shared infrastructure +-----+	
BaseEntity · GlobalExceptionHandler · ApiError	
ResourceNotFoundException · ConflictException	
BusinessException	

Packages:

- core/ - shared infra only (BaseEntity, exceptions, handler)
- modules/visitemanagement/ - Scheduling + Registration sub-modules
- Enums live in the module package (not core/)

2. Database Schema

Entity-Relationship Diagram

+-----+ time_slot +-----+			+-----+ visitor +-----+		
id	BIGINT	PK	id	BIGINT	PK
slot_date	DATE	NOT NULL	full_name	VARCHAR(150)	NOT NULL
start_time	TIME	NOT NULL	group_size	INT	NOT NULL, DEFAULT 1
end_time	TIME	NOT NULL	email	VARCHAR(200)	UNIQUE
max_capacity	INT	NOT NULL	phone	VARCHAR(30)	
status	VARCHAR(20)	NOT NULL	language	VARCHAR(40)	
guide_id	BIGINT		visitor_type	VARCHAR(20)	NOT NULL
created_at	INSTANT	NOT NULL	special_needs	VARCHAR(500)	
updated_at	INSTANT	NOT NULL	created_at	INSTANT	NOT NULL
+-----+			+-----+		
1			1		
*			1		
+-----+			+-----+		
registration					

```

+-----+
| id          BIGINT| PK                               |
| visitor_id   BIGINT| FK -> visitor (NOT NULL)                       |
| time_slot_id BIGINT| FK -> time_slot (nullable, NULL for          |
|              |      event-only registrations)              |
| event_id     BIGINT|                               |
| visit_purpose  VARCHAR(20)| NOT NULL                               |
| is_prospect  BOOLEAN| NOT NULL, DEFAULT false                     |
| status       VARCHAR(20)| NOT NULL, DEFAULT 'PENDING'                 |
| created_at   INSTANT| NOT NULL                               |
| updated_at   INSTANT| NOT NULL                               |
| UNIQUE (visitor_id, time_slot_id)              |
+-----+
| 1
|
| 1
+-----+
| briefing      |
+-----+
| id          BIGINT| PK                               |
| registration_id | FK -> registration              |
|              |      (NOT NULL, UNIQUE)        |
| delivered_at INSTANT|                               |
| staff_member VARCHAR(150)|                               |
| signature     VARCHAR(500)|                               |
| status        VARCHAR(20)| NOT NULL, 'PENDING'           |
| created_at    INSTANT| NOT NULL                       |
| updated_at    INSTANT| NOT NULL                       |
+-----+

```

registration.event_id -> event.id (no FK constraint, soft link)

```

+-----+
| event
+-----+
| id          BIGINT| PK                               |
| title       VARCHAR(150)| NOT NULL                       |
| event_type  VARCHAR(30)| NOT NULL                       |
| start_date_time TIMESTAMP| NOT NULL                       |
| end_date_time  TIMESTAMP| NOT NULL                       |
| max_capacity  INT| NOT NULL, DEFAULT 10          |
| status       VARCHAR(20)| NOT NULL, 'DRAFT'             |
| description   VARCHAR(1000)|                               |
| location     VARCHAR(150)|                               |
| created_at   INSTANT| NOT NULL                       |
| updated_at   INSTANT| NOT NULL                       |
+-----+

```

Table Summary

Table	Rows (design)	Purpose
`time_slot`	~500/year	Farm tour calendar slots (2/day max)
`visitor`	~2,000	Visitor database (individuals/groups)
`registration`	~4,000	Links visitor -> time slot **or** event
`briefing`	~3,000	Safety briefing per confirmed registration
`event`	~50/year	Open days, buyer visits, school/community days

Enums

Enum	Values	Table Column
`TimeSlotStatus`	`AVAILABLE`, `RESERVED`, `FULL`, `CANCELLED`, `COMPLETED`	`time_slot.status`
`RegistrationStatus`	`PENDING`, `CONFIRMED`, `REJECTED`, `CHECKED_IN`, `CANCELLED`	`registration.status`

```
| `BriefingStatus` | `PENDING`, `DONE` | `briefing.status` |
| `VisitorType` | `INDIVIDUAL`, `GROUP`, `SCHOOL`, `PARTNER` | `visitor.visitor_type` |
| `VisitPurpose` | `TOURISM`, `PURCHASE`, `PARTNERSHIP`, `INVESTMENT`, `EDUCATION`, `OTHER` |
`registration.visit_purpose` |
| `EventType` | `OPEN_DAY`, `PARTNER_BUYER`, `SCHOOL`, `COMMUNITY`, `OTHER` | `event.event_type` |
| `EventStatus` | `DRAFT`, `PUBLISHED`, `CANCELLED`, `COMPLETED` | `event.status` |
```

3. REST API Endpoints

Base URL: <http://localhost:8080/api/v1>

3.1 Time Slots - /api/v1/time-slots

Method	Path	Description	Status Codes
GET	/time-slots	List all slots (optional `?date=YYYY-MM-DD`)	200
GET	/time-slots/{id}	Get slot by ID	200, 404
POST	/time-slots	Create a new slot	201, 400, 409, 422
PUT	/time-slots/{id}	Update slot details	200, 404, 422
DELETE	/time-slots/{id}	Cancel a slot (soft-delete)	204, 404, 422
POST	/time-slots/{id}/assign-guide	Assign a guide (`?guideId=N`)	200, 404, 422
GET	/time-slots/availability?date=...	Availability view for a date	200

```
{
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "guideId": 5
}
```

Create/Update Request Body:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "booked": 3,
  "status": "RESERVED",
  "guideId": 5,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

TimeSlotResponse:

```
{
  "id": 1,
  "date": "2026-09-08",
  "startTime": "09:00",
  "endTime": "11:00",
  "maxCapacity": 10,
  "remaining": 7,
  "status": "RESERVED",
  "guideId": 5
}
```

AvailabilityResponse:

3.2 Visitors - /api/v1/visitors

Method	Path	Description	Status Codes
GET	/visitors	List all visitors	200
GET	/visitors/{id}	Get visitor by ID	200, 404
POST	/visitors	Create visitor	201, 400, 409

```
| `PUT` | `/visitors/{id}` | Update visitor | 200, 404, 409 |
```

```
{
  "fullName": "Alice Dupont",
  "groupSize": 2,
  "email": "alice@example.com",
  "phone": "+241 06 12 34 56",
  "language": "French",
  "type": "INDIVIDUAL",
  "specialNeeds": "Wheelchair access required"
}
```

VisitorRequest:

3.3 Registrations - /api/v1/registrations

Method	Path	Description	Status Codes
----- ----- ----- -----			
`GET`	`/registrations`	List (filter by `?timeSlotId=`, `?date=`, or `?prospect=true`)	200
`GET`	`/registrations/{id}`	Get registration by ID	200, 404
`POST`	`/registrations`	Create registration (visitor->slot, requires `visitPurpose`)	201, 400, 409, 422
`PATCH`	`/registrations/{id}/approve`	Approve -> triggers briefing creation	200, 404, 422
`PATCH`	`/registrations/{id}/reject`	Reject a pending registration	200, 404, 422
`PATCH`	`/registrations/{id}/check-in`	Check in a confirmed visitor	200, 404, 422
`PATCH`	`/registrations/{id}/cancel`	Cancel a registration	200, 404, 422

```
{
  "visitorId": 1,
  "timeSlotId": 10,
  "visitPurpose": "PURCHASE",
  "eventId": null
}
```

RegistrationRequest:

```
{
  "id": 1,
  "visitorId": 1,
  "visitorName": "Alice Dupont",
  "groupSize": 2,
  "timeSlotId": 10,
  "slotDate": "2026-09-08",
  "slotStart": "09:00",
  "eventId": null,
  "visitPurpose": "PURCHASE",
  "prospect": true,
  "status": "PENDING",
  "briefingId": null,
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

RegistrationResponse (new fields):

3.4 Briefings - /api/v1

Method	Path	Description	Status Codes
----- ----- ----- -----			
`GET`	`/registrations/{registrationId}/briefing`	Get briefing for registration	200, 404
`PATCH`	`/registrations/{registrationId}/briefing/deliver`	Mark briefing delivered	200, 422

```
{
  "staffMember": "Guard Camille",
  "signature": "digital-signature-or-name"
}
```

BriefingDeliverRequest:

3.5 Events - /api/v1/events

Method	Path	Description	Status Codes
GET	/events	List events (`?type=`, `?date=YYYY-MM-DD`)	200
GET	/events/{id}	Get event by ID	200, 404
POST	/events	Create event (default status `DRAFT`)	201, 400, 422
PUT	/events/{id}	Update event (not allowed when CANCELLED/COMPLETED)	200, 404, 422
DELETE	/events/{id}	Cancel event (soft-delete)	204, 404, 422
POST	/events/{id}/publish	Publish a DRAFT event	200, 404, 422
GET	/events/{id}/registrations	List registrations for the event	200, 404
POST	/events/{id}/register	Register a visitor for the event	201, 400, 404, 409, 422

```
{
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall"
}
```

EventRequest:

```
{
  "id": 1,
  "title": "Open Farm Day",
  "type": "OPEN_DAY",
  "startDateTime": "2026-10-04T10:00:00",
  "endDateTime": "2026-10-04T16:00:00",
  "maxCapacity": 60,
  "booked": 3,
  "status": "PUBLISHED",
  "description": "Discover the farm, meet the team, taste our products",
  "location": "Farm entrance hall",
  "createdAt": "2026-09-01T10:00:00Z",
  "updatedAt": "2026-09-01T10:00:00Z"
}
```

EventResponse:

```
{
  "visitorId": 1,
  "visitPurpose": "PURCHASE"
}
```

EventRegistrationRequest:

The registration created for an event has timeSlotId: null and carries the visitor's visitPurpose. Commercial purposes (PURCHASE, PARTNERSHIP, INVESTMENT) automatically mark the visitor as a prospect, exactly like time-slot registrations.

4. Business Rules

Scheduling Rules (SchedulingRules.java)

Rule	Value	Source
Max visitors per slot	**10**	Confirmed with module owners (Abdoul)
Max slots per day	**2**	Mockup: morning + afternoon
Closed day	**Sunday**	All module owners
Morning slot	09:00-11:00	Default
Afternoon slot	14:00-16:00	Default

Registration Rules

Rule	Enforcement
Cannot register on a CANCELLED slot	`BusinessRuleException`
No duplicate registration (visitor + slot)	`ConflictException` (409)
Group size must fit remaining capacity	`BusinessRuleException`
Only PENDING -> approve/reject	`BusinessRuleException`
Only CONFIRMED -> check-in	`BusinessRuleException`
Cannot cancel after CHECKED_IN	`BusinessRuleException`
CONFIRMED -> auto-creates `Briefing` (PENDING)	Service layer
Briefing delivery requires PENDING status	`BusinessRuleException`

Event Rules

Rule	Enforcement
`endTime` must be after `startTime`	`BusinessRuleException`
Only DRAFT events can be published	`BusinessRuleException`
Cannot publish an event that has already ended	`BusinessRuleException`
Cannot update CANCELLED or COMPLETED events	`BusinessRuleException`
Cannot cancel a COMPLETED event	`BusinessRuleException`
Registration only on PUBLISHED events	`BusinessRuleException`
No duplicate registration (visitor + event)	`ConflictException` (409)
`booked + groupSize > maxCapacity` -> reject	`BusinessRuleException` (422)
Booked count excludes `REJECTED` / `CANCELLED` registrations	Repository query

Visitor Rules

Rule	Enforcement
Duplicate email ->	`ConflictException` (409) On create and update
`@NotBlank` on fullName	Bean validation (400)
`@Email` on email	Bean validation (400)

Lead Generation (Prospect Detection)

The visitor management app doubles as a marketing tool for the Sales & Marketing team. When a visitor registers with a commercial purpose, the system automatically flags them as a prospect.

How it works:

1. During registration, the visitor must select a visitPurpose (mandatory field)
2. If the purpose is PURCHASE, PARTNERSHIP, or INVESTMENT, the system sets isProspect = true automatically
3. Sales & Marketing can query all prospects via GET /api/v1/registrations?prospect=true

VisitPurpose values:

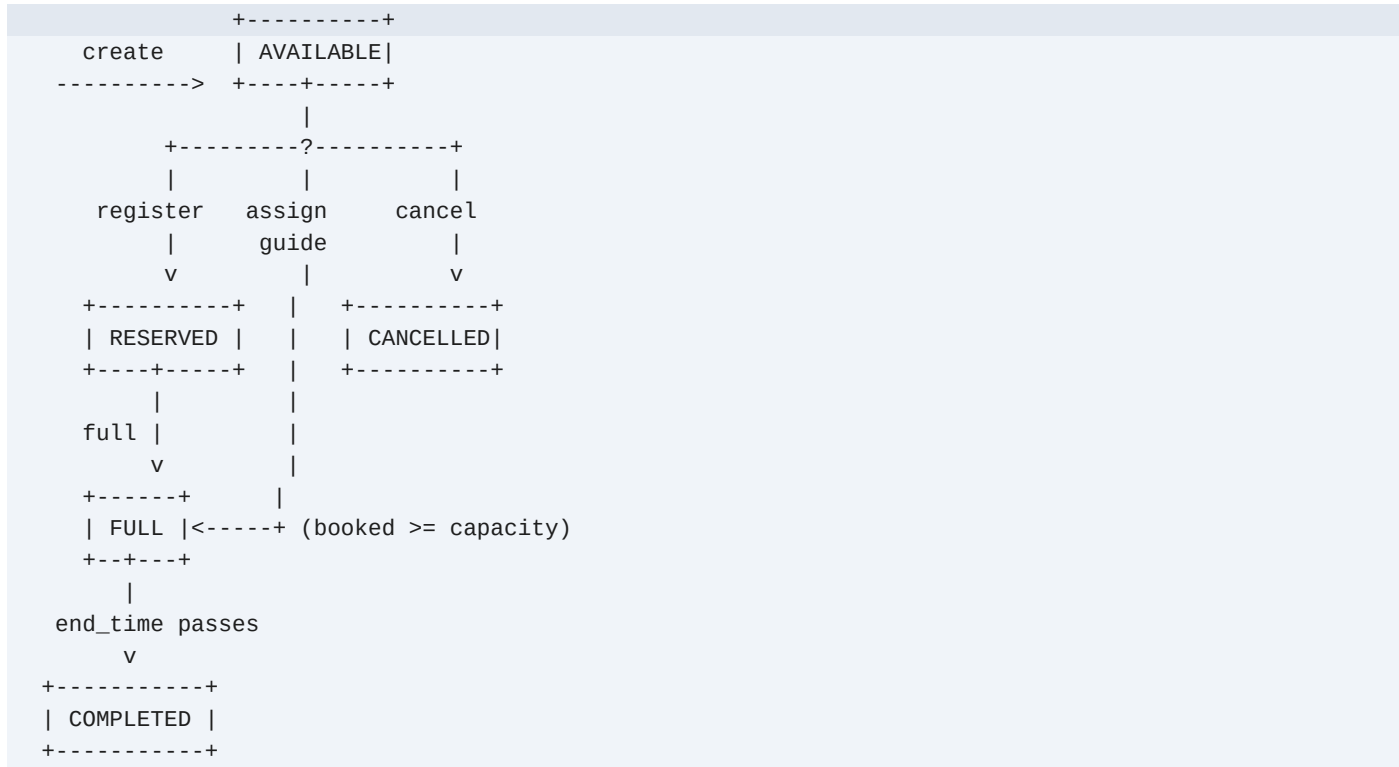
Purpose	Prospect?	Description
`TOURISM`	No	Leisure visit, tourism
`PURCHASE`	Yes	Interested in buying products
`PARTNERSHIP`	Yes	Business partnership opportunity
`INVESTMENT`	Yes	Investment opportunity
`EDUCATION`	No	Training, educational visit
`OTHER`	No	Other reasons

GET /api/v1/registrations?prospect=true

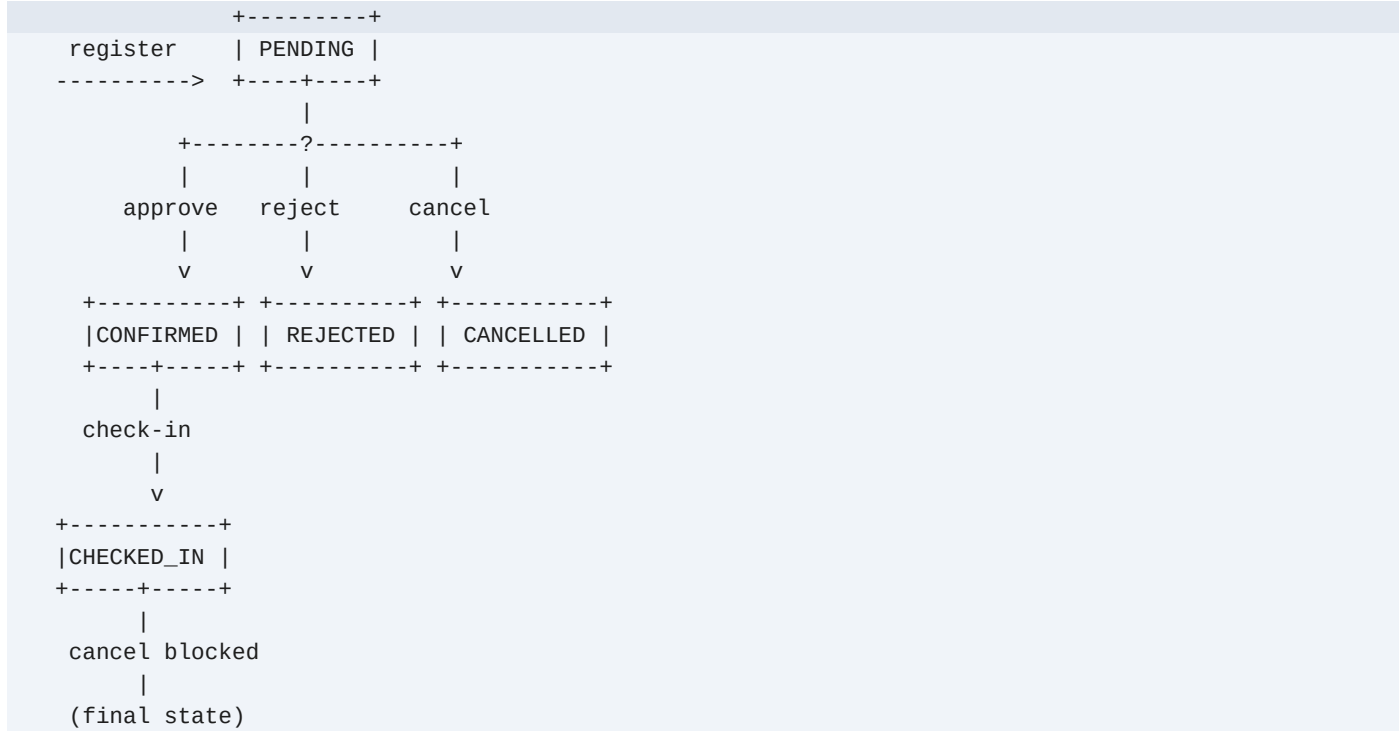
Prospect query example: Returns all registrations where isProspect = true, including visitor name, email, phone, and visit purpose.

5. Status Flow Diagrams

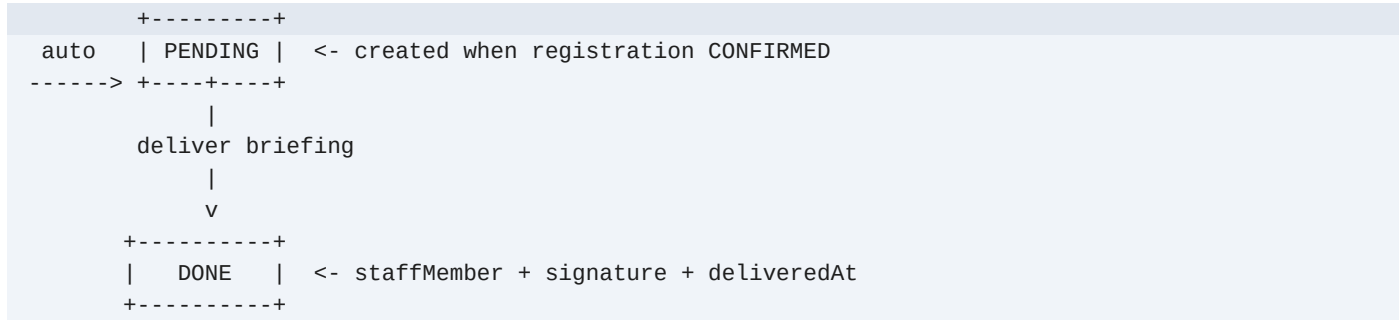
TimeSlot Status Flow



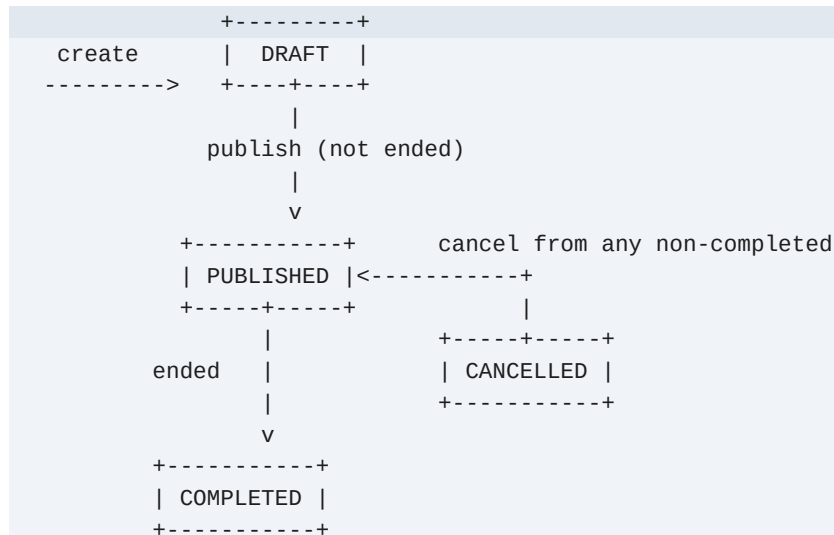
Registration Status Flow



Briefing Status Flow



Event Status Flow



6. Error Handling

All errors return a consistent JSON envelope via `GlobalExceptionHandler`:

```
{
  "status": 404,
  "error": "Not Found",
  "message": "Time slot 999 not found",
  "path": "/api/v1/time-slots/999"
}
```

Exception	HTTP Status	When
<code>`ResourceNotFoundException`</code>	404	Entity not found
<code>`ConflictException`</code>	409	Duplicate email, duplicate registration, overlapping slot
<code>`BusinessRuleException`</code>	422	Invalid state transition, capacity exceeded, Sunday creation
<code>`MethodArgumentNotValidException`</code>	400	Bean validation failures (<code>@NotBlank`</code> , <code>@Email`</code> , etc.)

7. Test Coverage Summary

Layer	Annotation	Tests	Framework
Repository	<code>`@DataJpaTest`</code> + H2	20	Spring Data + AssertJ
Service	<code>`@ExtendWith(MockitoExtension.class)`</code>	50	Mockito + AssertJ
Controller	<code>`@WebMvcTest`</code> + MockMvc	47	MockMvc + Mockito
Context	<code>`@SpringBootTest`</code>	1	Spring Boot
Total		**118**	

Test Files

```
src/test/java/com/infineonbit/sustainablefarm/
+-- SustainableFarmApplicationTests.java
+-- modules/visitormanagement/
    +-- repository/
        | +-- TimeSlotRepositoryTest.java      (5 tests)
        | +-- VisitorRepositoryTest.java       (4 tests)
        | +-- RegistrationRepositoryTest.java  (5 tests)
        | +-- BriefingRepositoryTest.java      (2 tests)
        | +-- EventRepositoryTest.java         (4 tests)
    +-- service/
        | +-- SchedulingServiceImplTest.java   (13 tests)
        | +-- RegistrationServiceImplTest.java (21 tests)
        | +-- EventServiceImplTest.java        (16 tests)
    +-- controller/
        | +-- TimeSlotControllerTest.java      (9 tests)
```



```

+-- VisitorControllerTest.java      (7 tests)
+-- RegistrationControllerTest.java  (12 tests)
+-- BriefingControllerTest.java     (5 tests)
+-- EventControllerTest.java        (13 tests)

```

8. Running the Backend

```

# Start PostgreSQL + Backend
docker compose up -d backend

# Backend available at
http://localhost:8080

# Swagger UI
http://localhost:8080/swagger-ui/index.html

# Actuator health
http://localhost:8080/actuator/health

# Run tests (Docker, no local JDK required)
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw test

# Compile only
docker run --rm -v "$PWD":/src -w /src \
  -v "$HOME/.m2":/root/.m2 \
  maven:3.9-eclipse-temurin-21 ./mvnw compile

```

9. Technology Stack

Component	Version	Notes
Java	21	LTS
Spring Boot	4.1.0	Modular monolith
Spring Data JPA	4.1.0	Derived queries + `@Query`
Hibernate ORM	7.4.1	`ddl-auto=update` (no Flyway)
PostgreSQL	17-alpine	Production database
H2	2.x	Test-only in-memory DB
SpringDoc OpenAPI	2.8.5	Swagger UI auto-generated
Lombok	1.18.46	Boilerplate reduction
Jackson	3.1.4	JSON serialization
Mockito	5.23.0	Service + controller tests
AssertJ	3.27.7	Fluent assertions

Last updated: 2026-09-09